

Aditya (Adi) Chinchure

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PhD Candidate in Computer Vision and NLP at UBC, building reasoning-inspired methods to **evaluate and improve vision-language and generative AI models**. My focus spans fine-grained image and video understanding & evaluation, surprise localization and error detection in videos, and improving the diversity and quality of image and video generation models.

EDUCATION

 University of British Columbia PhD in Computer Science <i>Supervisors: Dr. Leonid Sigal & Dr. Vered Shwartz</i> $\diamond$ Proposed Thesis: Reasoning-Inspired Methods for Evaluating and Improving Generative AI	Vancouver, BC May 2024 – Present
MSc, Computer Science , GPA: 4.0 (A+) <i>Supervisors: Dr. Leonid Sigal & Dr. Renjie Liao</i> $\diamond$ Thesis: Visual Question Answering with Contextualized Commonsense Knowledge	2024
BSc, Computer Science, Honors , GPA: 3.9 (A) $\diamond$ Thesis: Refinement Architectures for Referring Image Segmentation	2021

TECHNICAL SKILLS

Methods	VLM/LLM post-training (SFT, RLHF, GRPO), diffusion and flow-matching models, reward modeling, automatic & human evaluation, distributed multi-GPU training (DeepSpeed)
Stack	Python, PyTorch, HuggingFace (transformers, diffusers, TRL), vLLM, NumPy/pandas/sklearn, W&B, Slurm

RELEVANT WORK EXPERIENCE

 Ideogram <i>Machine Learning Research Intern Supervisor: Dr. Mohammad Norouzi</i> Core research team member for Ideogram 4.0 , ranked #1 open-weight model on DesignArena , with native layout specification. Designed 3+ VLM-based metrics and agentic pipelines for fine-grained text-to-image evaluation, RL, and system prompt optimization.	Nov 2025 – Mar 2026 Toronto, ON
 Toyota Technological Institute at Chicago <i>Visiting Student Researcher Supervisor: Dr. Matthew Turk</i> Built InterMit , a counterfactual method for intersectional bias mitigation in image generation: lower bias (0.33 vs. 0.52; higher diversity), in fewer steps (2.38 vs. 3.15), with better image quality than prior work.	Oct 2024 – Dec 2024 Chicago, IL, USA
 Borealis AI, Royal Bank of Canada <i>Machine Learning Research Intern Supervisors: Dr. Fred Tung, Dr. Leonid Sigal</i> Developed PD-EST , a process-disentangling transformer for event sequences such as financial transactions. Jointly learns a process mask and the sequence model, improving next-event time/type prediction with interpretable process decomposition.	Sep 2022 – May 2023 Vancouver, BC
 LEAP Project, UBC & BC Cancer Research <i>Research Assistant Supervisors: Dr. Ivan Beschastnikh, Dr. Aline Talhouk</i> Designed a module that enabled 2x faster data retrieval from REDCap research databases for a privacy-focused federated ML project.	May 2020 – Aug 2020 Vancouver, BC
 Hypercontext <i>ML Engineer (Co-op)</i> Developed BERT models for text classification, sentiment analysis and entity recognition. Orchestrated an end-to-end pipeline including data cleanup, a Flask REST API for serving, and retraining and deployment with Docker on AWS.	May 2019 – Aug 2019 Toronto, ON

PUBLICATIONS

Spotlight: Identifying and Localizing Video Generation Errors Using VLMs Aditya Chinchure , Sahithya Ravi, Pushkar Shukla, Vered Shwartz, Leonid Sigal <i>European Conference on Computer Vision, ECCV 2026</i> Paper Project $\diamond$ 1,600+ annotated video generation errors, including temporal localization, on videos from Veo 3, LTX-2 and Seedance (700+ downloads on Hugging Face in 2 months). VLM-based, inference-time, and multi-agent strategies for error detection and localization.
SPIKE-RL: Video-LLMs meet Bayesian Surprise Sahithya Ravi, Aditya Chinchure , Raymond Ng, Leonid Sigal, Vered Shwartz <i>International Conference on Learning Representations, ICLR 2026</i> Paper Project $\diamond$ Bayesian-surprise frame sampling and GRPO-based belief optimization for Video-LLMs; improved performance over uniform sampling on five downstream benchmarks.

Mitigate One, Skew Another? Tackling Intersectional Biases in Text-to-Image Models

Pushkar Shukla*, **Aditya Chinchure***, Emily Diana, Alexander Tolbert, Kartik Hosanagar, Vineeth N Balasubramanian, Leonid Sigal, Matthew A. Turk

Empirical Methods in Natural Language Processing, Findings, EMNLP 2025 | [Paper](#)

- ◊ Introduces BiasConnect, the first intersectional bias analysis in T2I models using concept-level counterfactuals, and InterMit, which uses intersectionality to achieve stronger bias mitigation with fewer steps.

Black Swan: Abductive and Defeasible Video Reasoning in Unpredictable Events

Aditya Chinchure*, Sahithya Ravi*, Raymond Ng, Vered Shwartz, Boyang Li, Leonid Sigal

Computer Vision and Pattern Recognition, CVPR 2025 | [Paper](#) | [Project & Leaderboard](#)

- ◊ Video reasoning benchmark spanning 1,655 videos and 15,400+ questions testing explanation of unexpected events and belief revision. Led to the Black Swan Challenge @ CogVL Workshop, CVPR 2026.

From Local Concepts to Universals: Evaluating the Multicultural Understanding of Vision-Language Models

Mehar Bhatia, Sahithya Ravi*, **Aditya Chinchure***, Eunjeong Hwang, Vered Shwartz

Empirical Methods in Natural Language Processing, EMNLP 2024 | [Paper](#) | [Project](#)

- ◊ Evaluating multicultural understanding in VLMs: cross-cultural image retrieval across 50 countries and culture-specific visual grounding across 15, exposing wide performance gaps and cultural bias.

TIBET: Identifying and Evaluating Biases in Text-to-Image Generative Models

Aditya Chinchure*, Pushkar Shukla*, Gaurav Bhatt, Kiri Salij, Kartik Hosanagar, Leonid Sigal, Matthew A. Turk

European Conference on Computer Vision, ECCV 2024 | [Paper](#) | [Project](#)

- ◊ Automatic, prompt-specific bias quantification for image generation using counterfactual reasoning. Works with any black-box T2I model, and achieves concept-level explainability.

VLC-BERT: Visual Question Answering with Contextualized Commonsense Knowledge

Sahithya Ravi*, **Aditya Chinchure***, Leonid Sigal, Renjie Liao, Vered Shwartz

Winter Conference on Applications of Computer Vision, WACV 2023 | [Paper](#) | [Code](#)

- ◊ Vision-language-commonsense transformer that generates, selects and encodes question-specific contextual knowledge from COMET, outperforming static knowledge-base models on OK-VQA and A-OKVQA.

* indicates equal contribution

OTHER PUBLICATIONS

Position: World Models must live in Parallel Worlds. Sahithya Ravi*, **Aditya Chinchure***, Pushkar Shukla, Vered Shwartz, Leonid Sigal. *NeurIPS Workshop (LAW), NeurIPS-W 2025* | [Paper](#)

PD-EST: Process-disentangling Event Sequence Transformer. **Aditya Chinchure**, Fred Tung, Leonid Sigal. *Borealis AI Report*

LEAP: Private and Federated Data Analysis for Healthcare. Matheus Stolet, Chris Yoon, Kalli Leung, **Aditya Chinchure**, Mathias Lécuyer, Aline Talhouk, Ivan Beschastnikh. *Poster, BC's AI Showcase, UBC CAIDA* | [Project](#)

AWARDS, SCHOLARSHIPS, AND GRANTS

CIFAR AI Safety Solution Network on Synthetic AI, Research Grant, UofT Sep 2026 – Present
Grant for developing methods for deepfake detection in videos. \$12,654 per year.

Four Year Doctoral Fellowship, UBC May 2024 – Present
For outstanding students joining UBC for their doctoral studies. \$24,000 per year + tuition.

Vector Institute Research Grant 2022 – Present
Yearly research grant (\$6,000) for graduate students affiliated with the Vector Institute for AI.

Nominated for CRA Outstanding Undergraduate Researcher Award Oct 2020
Nominated by Dr. Ivan Beschastnikh for my work on the LEAP project.

Faculty of Science – International Student Scholarship, UBC Sep 2019
A monetary award of \$10,000 for strong academic achievement in computer science.

SERVICE, TEACHING, AND OTHER ACTIVITIES

Organizing: Lead organizer, [CogVL workshop](#) at CVPR 2026, including the [Black Swan Challenge](#).

Mentorship & Talks: Mentoring three undergrad and master's students on multimodal projects. Talks at TU Delft and CIFAR AI Safety Solution Network on Synthetic AI (2026); guest lectures in CPSC 532V: Commonsense Reasoning (2025–26).

Teaching Assistant: Computer Vision (CPSC 425), Intro to AI (CPSC 322), and three database courses at UBC.

Reviewing: CVPR*, ECCV*, ICCV, NeurIPS, AAAI, TPAMI and workshops at CVPR and ICCV (2022–2026). **Outstanding reviewer designation at ECCV 2024 and CVPR 2025*.

Photography: 260M+ views on [Unsplash](#); professional commercial work. Informs my research on image & video generation.